

PRT661 – Data Science Practice

Assessment 1: Project Proposal and Design

Deep Learning-Based Classification of Early and Late Blight in Potato Leaves

(Theme 2 – Predictive Analytics and Forecasting)

Danala Group 7	
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GitHub Repository Link: <https://github.com/mahdihasan0017/PRT661-Data-Science-Practice-Dan-7-Theme-2>

Course Code: PRT661

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1. Project Overview and Objectives

Plant diseases can reduce crop quality and productivity when visible symptoms are not recognised promptly. Manual inspection is useful but may be time-consuming and dependent on the observer's experience. This project proposes a simple binary image-classification system that distinguishes potato leaves affected by Early Blight from those affected by Late Blight. The project will use the public PlantVillage image dataset and develop a convolutional neural network (CNN) to learn disease-related visual patterns. The intended output is a reproducible proof-of-concept decision-support tool rather than a replacement for professional crop diagnosis. The project objectives are to:

- acquire, organise and document a suitable public image dataset;
- preprocess potato-leaf images into a consistent modelling format;
- develop a CNN for Late Blight versus Early Blight classification;
- evaluate predictive performance using multiple binary-classification metrics; and
- visualise results and critically assess limitations, risks, ethics, privacy and security.

2. Theme Justification

The project fits Theme 2 Predictive Analytics and Forecasting because labelled images are used to train a model that predicts the class of a new image. The target is binary: Late Blight or Early Blight. It also covers data acquisition, storage, processing, machine-learning analytics and visualisation.

3. Problem Statement

Early Blight and Late Blight are two important potato diseases that can produce visually similar symptoms on leaves, making accurate differentiation challenging when relying only on manual inspection. Recognition may also vary with image quality, symptom severity, and observer experience. The research question for this project is:

Can a CNN accurately distinguish potato leaves affected by Early Blight from those affected by Late Blight using visible image characteristics?

The project will focus only on binary disease classification and performance evaluation. It will not provide treatment, pesticide, or disease-management recommendations.

4. Data Source and Initial System Architecture

PlantVillage contains 54,306 leaf images across 38 crop-disease classes. This project will use only Potato_Early_blight and Potato_Late_blight, with 1,000 images in each class, giving a balanced dataset of 2,000 images. Images will be stored in structured folders, while GitHub will maintain code, documentation, and dataset links.

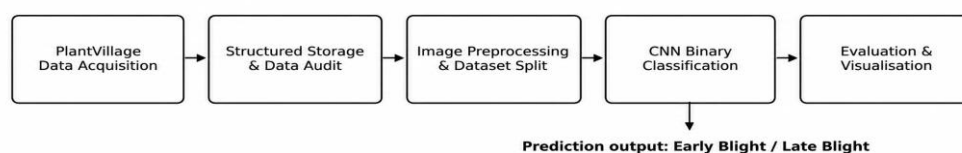


Figure 1. Initial architecture for the proposed binary leaf-disease classification system.

5. Workflow and Analytical Plan

The project will use Python with PyTorch, NumPy, Scikit-learn and Matplotlib. Image files and labels will first be checked and summarised. Images will be resized to 224×224 pixels and normalised. Horizontal flipping, small rotations and limited zoom will be applied only to training images. A stratified train-validation-test split will preserve class proportions, while a leaf-group-aware split will be preferred when available to reduce leakage.

A lightweight CNN will be the primary model, using convolution and pooling layers followed by dense layers and a sigmoid output. Class weighting will address imbalance. Performance will be reported using accuracy, precision, recall, F1-score, ROC-AUC and a confusion matrix. Early-Blight recall is important because a false negative is a diseased leaf predicted as healthy. Visualisations will include class distribution, sample images, learning curves, confusion matrix, ROC curve and prediction examples.

Table 1. Proposed workflow and milestones.

Stage	Key activity	Planned output
1	Acquire and audit data	Dataset inventory and class summary
2	Preprocess and split images	Reproducible train/validation/test sets
3	Develop and train CNN	Saved model and training history
4	Evaluate predictions	Metrics, confusion matrix and ROC curve
5	Interpret and document	Figures, limitations and final report materials

6. Project Management and Task Allocation

A shared GitHub repository will support version control, code review and documentation. GitHub Projects will track tasks, milestones, ownership, progress, completion, the project timeline and scope changes using a Backlog–In Progress–Review–Done workflow. Members will have primary responsibilities but will review each other’s work. Commits and board activity will provide collaboration evidence.

Table 2. Proposed team structure and responsibilities.

Team member	Primary responsibility
Member 1	Data acquisition and repository setup
Member 2	Image preprocessing and exploratory analysis
Member 3	CNN development and model training
Member 4	Evaluation, visualization and error analysis
Member 5	Documentation, ethics review and final integration

7. Initial Risk Analysis

Table 3. Initial project risks and mitigation strategies.

Risk	Potential impact	Mitigation
Visual similarity between diseases	Misclassification between Early Blight and Late Blight	Use image augmentation and class-specific evaluation metrics.
Overfitting	Poor generalization	Use augmentation, early stopping and regularization.
Data leakage	Inflated performance	Split data before augmentation and verify no overlap between sets.
Limited computing resources	Slow experiments	Use 224×224 images and a lightweight CNN.
Domain shift	Lower accuracy on real field images	Clearly state dataset limitations and recommend external field validation.
Version/data loss	Inconsistent team outputs	Use GitHub branches, regular commits and protected shared storage.

8. Ethics, Privacy and Security Considerations

The dataset contains plant images rather than identifiable human information, so privacy risk is low. The team will still respect licensing and attribution, maintain data provenance, control access to shared files and keep backups. The main ethical risk is overstating model capability. PlantVillage images were collected under controlled conditions, so performance may fall with field lighting, clutter, orientation changes or similar symptoms (Mohanty et al., 2016). Results will therefore be reported as experimental evidence only. The system will not recommend pesticides or treatments, and errors, class imbalance and limitations will be reported transparently. Future deployment would require independent field testing and expert agricultural oversight.

References

- Hughes, D. P., & Salathé, M. (2015). An open access repository of images on plant health to enable the development of mobile disease diagnostics. arXiv. [Link](#)
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521, 436–444. [Link](#)
- Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science*, 7, 1419. [Link](#)
- Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data*, 6, 60. [Link](#)
- Project-AgML. (n.d.). PlantVillage classification dataset. Hugging Face Datasets. [Link](#)
- PlantVillage dataset repository. (n.d.). PlantVillage deep learning paper dataset. [Link](#)
- Dataset access:** [PlantVillage image dataset](#)

Section 2: Project Update Summary

Danala Group 7 | PRT661 – Data Science Practice | Semester 2, 2026

Since Assessment 1, the project has progressed from a proposed binary CNN to an implemented comparison of logistic regression, PotatoCNN and MobileNetV2. The workflow now includes leaf-group-aware splitting, validation-selected thresholds, internal and external evaluation, and reliability analysis.

Changes since Assessment 1

Area	Previous state	Current state	Justification
Data and split	PlantVillage; grouped splitting preferred	2,000 images; 500 leaf groups; zero shared groups	Reduce leakage between related leaf images.
Models	Lightweight CNN proposed	Logistic regression, PotatoCNN and MobileNetV2	Compare traditional, custom and transfer-learning models.
Evaluation	Core metrics and visualisations planned	Locked internal test, PlantDoc evaluation and diagnostics	Assess generalization and reliability.
Delivery	Proposal and planned repository	Executed notebook, saved models, reports and repository evidence	Provide traceable implementation and results.

Contribution summary

Team member / student ID	Documented contribution area
Kavya Sree Sunkara / S386025	Data acquisition and repository setup
Sayed Mahidul Islam Rafel / S387949	Image preprocessing and exploratory analysis
Mahdi Hasan Abdullah / S394885	Model development and training
Partha Mallik / S392819	Evaluation, visualisation, error analysis and evidence uploads
Datta Sai Kanaparthi / S385673	Documentation, ethics review and final integration

Evidence locations

Evidence	Location / supporting record
Implementation	notebooks/CODE.ipynb in the accompanying repository package
Results and figures	artifacts/tables/ ; artifacts/figures/
Architecture and planning	architecture/ ; planning/
Project governance	project_management/ ; commit history

[GitHub repository: Danala Group 7](#)

MobileNetV2 achieved 99.67% accuracy on the internal test and 58.56% on PlantDoc. The external performance gap supports retaining the research proof-of-concept scope and prioritising representative field data and further validation.

Section 3: Assessment 2 Technical Development

3.1 Continuity with Assessment 1 and evidence basis

The original research question asks whether a CNN can distinguish Early Blight from Late Blight using visible potato-leaf characteristics. The implemented work retains the same two-class objective and adds stronger controls, comparators and external evaluation. The current milestone demonstrates a working analytical pipeline and exposes limitations that must guide the next stage.

Section 1 now preserves the complete four-page Dan Group 7 Theme 2 PP.pdf supplied in this update. Its cover identifies Danala Group 7, all five students, Semester 2 of 2026, a submission date of 11 August 2026 and the GitHub repository. This team-specific proposal replaces the earlier placeholder copy. Its existing wording and original figure remain unchanged.

What has changed since the proposal

The proposal already preferred leaf-group-aware splitting when possible. The implementation makes it mandatory and validates zero shared groups. The proposal's CNN concept is now accompanied by a traditional logistic model and trained MobileNetV2 comparator. Validation controls checkpoints and operating thresholds; a separate PlantDoc corpus tests source-domain transfer. Calibration, segmentation, perturbation, ablation and explainability outputs broaden the evidence.

The original proposal incorrectly describes a false negative as a diseased leaf predicted healthy. There is no healthy class in this task. Throughout this continuation, Late Blight is label 1 and Early Blight is label 0: a false negative is Late Blight predicted Early Blight. This correction is documented here while preserving the supplied Assessment 1 pages.

Review method and source precedence

This report is based on inspection of the executed notebook, exported CSVs, packaged model files, planning records and existing reports. No model was retrained for this progress review. Saved outputs establish evidence of a previous run; independent rerun reproducibility remains to be demonstrated. Where narrative or slides conflict with current numerical outputs, this report uses the notebook and matching CSVs.

Source	Use in this progress report
Assessment brief, pages 1–5	Section order, searchable PDF, page/font limits and individual requirements.
Dan Group 7 Theme 2 PP.pdf	Team-specific Assessment 1 baseline, identities, IDs, date and repository URL.
Notebook and artifacts/tables/	Implemented method and numerical evidence.
Planning records and ZIP inventory	Status, governance gaps and deliverable availability.

Primary implementation:

notebooks/Potato_Blight_Final_Feedback_Incorporated_Colab.ipynb. This report references notebook cell headings rather than unstable execution counters.

New source records: Dan Group 7 Theme 2 PP.pdf; team_structure_responsibilities.pdf; CODE.ipynb.

3.2 Updated system architecture

From a single-model proposal to a controlled comparison pipeline

The implementation separates data preparation, fitting, validation decisions and final evaluation. CSV outputs preserve the connection between labels, split membership, predictions and reported metrics. Raw source revisions are pinned, and derived transformations are kept distinct from the original images.

Stage	Implemented component and output
1. Acquire	Pinned PlantVillage and PlantDoc sources; inventory and provenance.
2. Audit and partition	Official leaf-group identifiers; 350/75/75 train/validation/test groups; overlap audits.
3. Prepare	224×224 RGB and ImageNet normalisation for neural models; 32×32 grayscale for logistic regression; augmentation only in training.
4. Fit	Logistic pipeline, compact PotatoCNN and ImageNet-pretrained MobileNetV2.
5. Validate	Best checkpoint and model-specific F1 threshold selected on validation data.
6. Evaluate	Frozen internal test; unchanged models and thresholds on external PlantDoc.
7. Diagnose	Errors, Grad-CAM, features, calibration, ablations, segmentation and perturbations.
8. Package and present	CSV/PNG evidence, saved models, environment record and optional Gradio probability display.

Component boundaries

The loader supplies images and labels; training updates parameters; validation selects checkpoints and operating policies; evaluation exports predictions and metrics. PlantDoc labels are reserved for evaluation. The model and its preprocessing and threshold must be treated as one versioned configuration, because changing representation changes the probability scale.

Design choices and tradeoffs

Python, PyTorch and Torchvision support the implemented neural workflow, while scikit-learn provides the traditional baseline and metric routines. A file-based design suits the 2,000-image

experimental corpus and supports direct inspection. A database, streaming pipeline and production hosting are not demonstrated by the supplied work. The architecture documents describe deployment possibilities, but these remain design extensions.

Design evidence: architecture/02_final_high_level_system_architecture.png;
architecture/03_final_workflow_diagram.png;
architecture/ARCHITECTURE_AND_DESIGN.md;
artifacts/documentation/SYSTEM_ARCHITECTURE.txt. The table above is a readable rendering of the implemented flow.

3.2a Architecture figure

Implemented separation of data, fitting, validation and evaluation

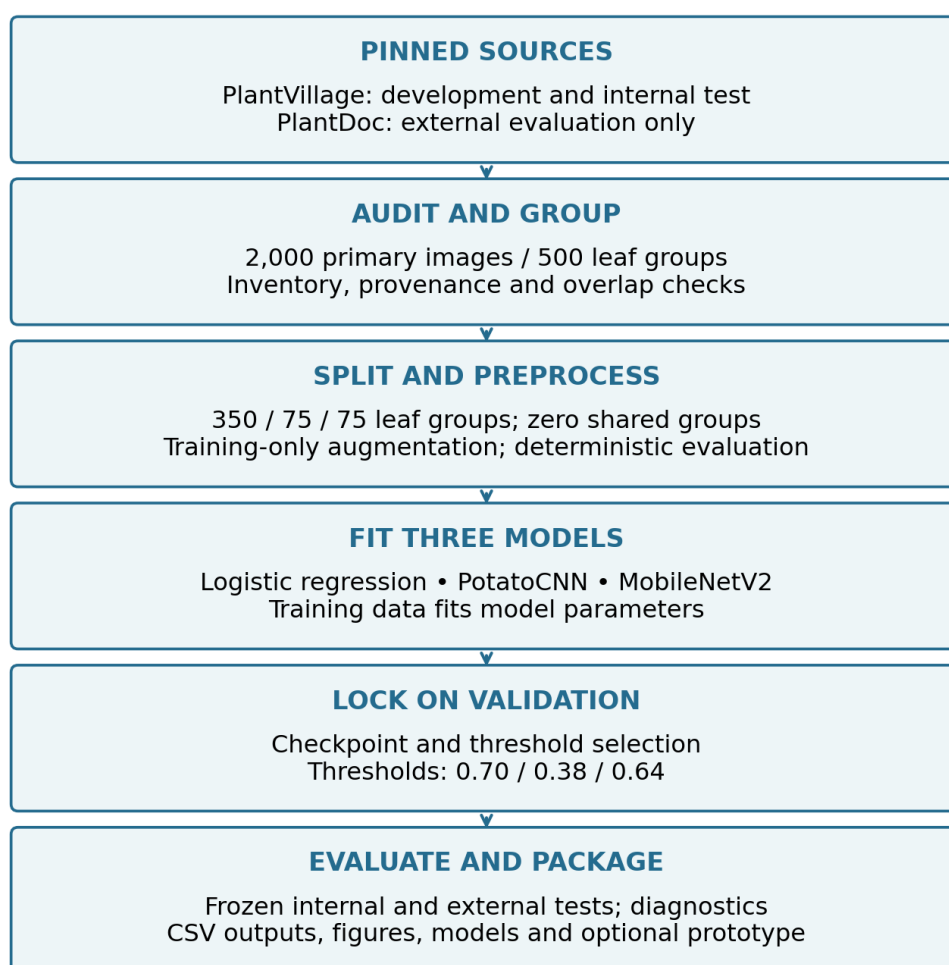


Figure 2: Updated architecture, redrawn from the supplied implemented-system specification. PlantDoc remains external to fitting and threshold selection.

Source architecture specification

The validation gate is the key control: each model is frozen before the internal and external evaluations. Augmentation operates only on the training partition; diagnostics do not retrospectively redefine the headline result.

3.3 Data acquisition, provenance and storage

Pinned sources and traceable experimental inputs

The recorded PlantVillage potato subset contains 2,000 JPEG images at 256×256 pixels, with 1,000 images per disease class and an aggregate size of approximately 34.71 MiB. The notebook resolves 500 underlying leaf groups with four captures per group. This distinction matters: 2,000 image files do not represent 2,000 independent specimens.

Source	Recorded revision / use
PlantVillage	Training, validation and internal testing.
PlantDoc	222 external images: 117 Early Blight and 105 Late Blight; no fitting or tuning.

Storage organisation

The notebook uses a Colab working directory under /content/PlantVillage-Dataset/raw and produces tables, figures, models and documentation. The supplied Windows repository contains the notebook, assessment reports, architecture and planning material, and exported tables/figures. Absolute Colab paths in CSVs describe the original runtime and will need reconstruction or mapping when reproducing the work elsewhere.

Three trained model files are present inside Potato_Blight_Final_Report_Ready.zip: models/baseline_logistic.joblib, models/best_potato_cnn.pt and models/best_mobilenet_v2.pt. They are not present under the extracted repository's artifacts/models/ location, although the task register points there. Model creation is evidenced by the archive, but repository packaging is incomplete.

Five-V profile

Dimension	Project-specific interpretation
Volume	Small, auditable image corpus; 2,000 primary images / 500 leaf groups.
Velocity	Static revision-pinned sources; no live ingestion demonstrated.
Variety	RGB captures, segmented representations and a different external image source.
Veracity	Public labels, group identifiers and hash audits; no claim of expert relabelling.
Value	Reproducible comparison and a preliminary two-disease classification prototype.

The supplied provenance note flags that PlantVillage redistribution terms require confirmation and records PlantDoc as CC BY 4.0. These are the project’s recorded licensing observations, not a fresh legal verification. Retain source attribution and resolve redistribution permissions before including raw images in a public release.

artifacts/tables/five_vs_summary.csv;

artifacts/documentation/DATA_PROVENANCE_AND_LICENSES.txt;

Potato_Blight_Final_Report_Ready.zip (workspace root).

3.4 Data engineering, cleaning and leakage control

Partition by underlying leaf before any augmentation

The notebook inventories file properties and class labels, resolves official leaf identifiers and checks the expected group structure. Each leaf group is assigned wholly to one partition. Class-wise group allocation gives a balanced training set and slightly different validation/test class totals because groups, rather than individual images, are allocated.

Partition	Early	Late	Images	Leaf groups
Training	700	700	1,400	350
Validation	148	152	300	75
Test	152	148	300	75
Total	1,000	1,000	2,000	500

Recorded leakage checks

Comparison	Shared groups	Exact matches	Near matches
Train–validation	0	0	0
Train–test	0	0	0
Validation–test	0	0	0

The image audit checks MD5 equality and perceptual dHash similarity at Hamming distance ≤ 4 . The PlantDoc overlap export records zero exact matches and zero external images with a near-match to the primary corpus at that threshold. These controls address the measured overlap routes; they do not prove that public labels are correct or that no other dependence exists.

Preprocessing contract

Neural inputs are converted to RGB, resized to 224×224 and normalised with ImageNet statistics. Training adds horizontal flipping, rotation up to 15 degrees and random resized crop with scale 0.9–1.0. Validation and test preprocessing is deterministic. Logistic regression uses 32×32 grayscale pixels with a scaler fitted within the training pipeline.

The balanced 700/700 training labels justify disabling class weighting. No oversampling or synthetic image creation is needed to correct class imbalance in this run. Segmented images

are handled as a separate sensitivity experiment and must not be mixed into the primary colour evaluation without an explicitly validated policy.

artifacts/tables/split_summary.csv; train_split.csv; validation_split.csv; test_split.csv; leaf_group_overlap_audit.csv; duplicate_audit.csv; plantdoc_external_overlap_audit.csv. Notebook headings: official leaf-group-aware split; dataloaders; cross-split duplicate audit.

3.5 Analytics and model implementation

Three completed modelling approaches

Model	Implementation	Purpose
Logistic regression	32 × 32 grayscale; StandardScaler; liblinear solver; maximum 1,500 iterations; seed 42.	Traditional reference for the neural models.
PotatoCNN	Three 3 × 3 convolution blocks with 32/64/128 channels, ReLU and max pooling; dropout 0.30; one output logit; 93,377 parameters.	Compact task-specific model.
MobileNetV2	ImageNet-pretrained weights; binary head; head-training phase followed by partial feature fine-tuning.	Transfer-learning comparator.

Custom CNN optimisation

PotatoCNN uses BCEWithLogitsLoss and Adam at a learning rate of 0.001, with batch size 32 and a maximum of 20 epochs. Early stopping monitors validation loss with a patience of five epochs and restores the best weights. Sigmoid converts the output logit to a Late-Blight score for evaluation. Balanced training labels leave the positive-class loss weight disabled.

Transfer-learning optimisation

The MobileNetV2 classifier is replaced with a single-output linear head. Feature parameters are initially frozen; the head phase permits up to four epochs at a learning rate of 0.001. The final two feature blocks are then made trainable for up to four epochs at 0.0001. The best validation-loss checkpoint is restored before threshold selection. The saved history supplies evidence of the executed stages.

Decision policy

Model	Validation-selected threshold
Logistic regression	0.70
PotatoCNN	0.38
MobileNetV2	0.64

The shared threshold-selection function optimises validation F1. A Late-Blight prediction occurs when its score meets the model's threshold. These are model-specific operating policies, not interchangeable probability cutoffs. The external evaluation applies the same frozen thresholds without using PlantDoc labels for selection.

artifacts/tables/custom_cnn_configuration.csv; training_history.csv; mobilenet_v2_training_history.csv; final_model_comparison.csv. Models are in the report-ready ZIP. Notebook headings: Traditional baseline; Lightweight CNN; MobileNetV2 transfer-learning comparator.

3.5a Data split and training figures

Balanced classes, independent leaf groups and recorded optimisation

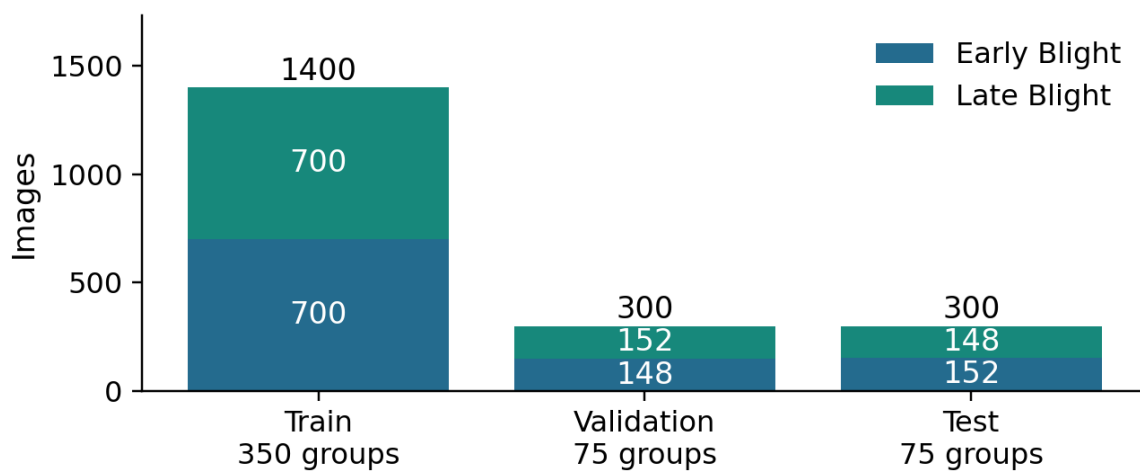


Figure 3. Image counts by class and split, with the number of underlying leaf groups below each partition. Source: split_summary.csv.

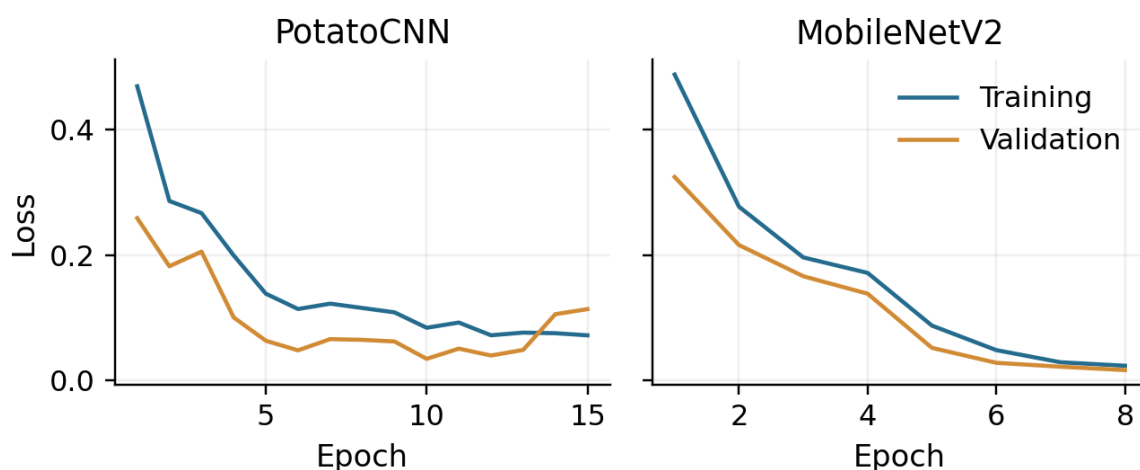


Figure 4. Recorded training and validation losses for PotatoCNN and MobileNetV2. These show saved optimisation histories, not an independent rerun.

Split evidence

PotatoCNN training history

MobileNetV2 training history

3.6 Locked internal evaluation

300 test images from 75 previously unseen leaf groups

All models are evaluated on the same internal test partition, comprising 152 Early-Blight and 148 Late-Blight images. Late Blight is the positive class. Accuracy measures overall correctness; precision and recall describe Late-Blight predictions; F1 balances their tradeoff; ROC-AUC measures score ranking across thresholds.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic	0.9100	0.8903	0.9324	0.9109	0.9624
PotatoCNN	0.9533	0.9408	0.9662	0.9533	0.9952
MobileNetV2	0.9967	1.0000	0.9932	0.9966	1.0000

Model	TN	FP	FN	TP	Errors
Logistic	135	17	10	138	27
PotatoCNN	143	9	5	143	14
MobileNetV2	152	0	1	147	1

MobileNetV2 correctly classifies 299 of 300 internal images. The custom model improves accuracy by 4.33 percentage points over logistic regression; MobileNetV2 improves by another 4.33 points over the custom model. These are controlled-source results and should be interpreted together with the external test, not used as a stand-alone field-performance claim.

Paired comparison and statistical limits

Comparison	Exact McNemar p-value
Logistic vs PotatoCNN	0.02412
Logistic vs MobileNetV2	0.0000002161
PotatoCNN vs MobileNetV2	0.0009766

The stored image-level McNemar tests report nominal p-values below 0.05 for all pairs. However, each test leaf contributes four related captures, so image-level independence is questionable. Multiple pairwise comparisons also warrant care. Treat these tests as exploratory support; leaf-group-aware uncertainty estimates and repeated-seed evaluation remain useful follow-up work.

Threshold correction

At threshold 0.50, PotatoCNN also has accuracy 0.9533, but Late recall is 0.9527. At 0.38, recall rises to 0.9662 and precision falls to 0.9408. Equal accuracy therefore does not mean identical predictions. The continuation corrects the older narrative implying that decisions were unchanged.

artifacts/tables/final_model_comparison.csv; model_mcnemar_tests.csv;
validation_threshold_comparison.csv; misclassified_samples.csv. Counts and rates shown here are taken from the saved exports.

3.7 External evaluation and generalisation

Completed PlantDoc experiment reveals substantial domain shift

The supplied notebook and external CSVs demonstrate that PlantDoc evaluation has already been performed. This supersedes the earlier progress slides, which list external validation as future work. The external cohort contains 222 images with 117 Early-Blight and 105 Late-Blight labels; it was not used to fit parameters or select thresholds.

Model	Accuracy	Late recall	Late F1	ROC-AUC
Logistic	0.4865	0.5524	0.5043	0.5044
PotatoCNN	0.5135	0.0762	0.1290	0.5361
MobileNetV2	0.5856	0.4095	0.4831	0.6210

Model	Internal accuracy	External accuracy	Drop, percentage points
Logistic	91.00%	48.65%	42.35
PotatoCNN	95.33%	51.35%	43.98
MobileNetV2	99.67%	58.56%	41.11

MobileNetV2 has the highest external accuracy and ROC-AUC among the three models, but logistic regression has the highest external Late-Blight F1. This illustrates why “best” depends on the metric. MobileNetV2 misses 62 of 105 Late-Blight images; PotatoCNN misses 97. Internal success does not transfer reliably to this source.

What the result establishes

The experiment provides direct evidence of cross-source performance degradation under the frozen evaluation policy. The zero-overlap audit reduces the risk that external scores are inflated by matched images. The result justifies retaining the proof-of-concept scope and prioritising representative data over further optimisation of the controlled benchmark.

What remains uncertain

PlantDoc is a small, public, internet-sourced dataset, not a prospective farm study with independently verified diagnoses. Differences may involve capture conditions, background, symptom presentation or label quality. The current evidence does not isolate their causal

contributions. Future work should use specimen-grouped, expert-reviewed data across sites and devices, while retaining an untouched confirmation cohort.

artifacts/tables/plantdoc_external_metrics.csv; plantdoc_external_predictions.csv;
 plantdoc_external_overlap_audit.csv. Supporting figure:
 artifacts/figures/plantdoc_external_confusion_matrices.png.

3.7a Internal and external performance figures

The external source exposes limits concealed by high internal accuracy

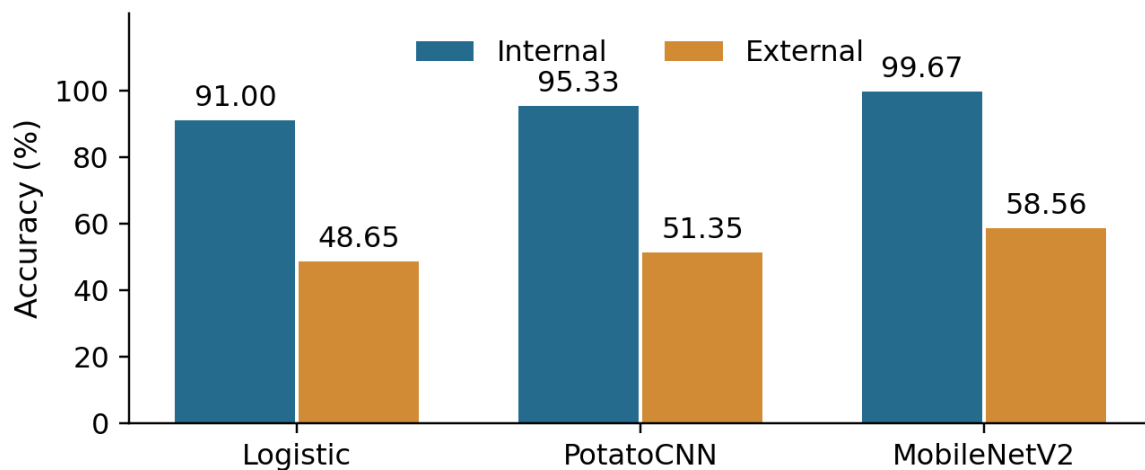


Figure 5. Accuracy on the 300-image internal test and 222-image external PlantDoc set. Values are percentages; frozen model-specific thresholds are retained.

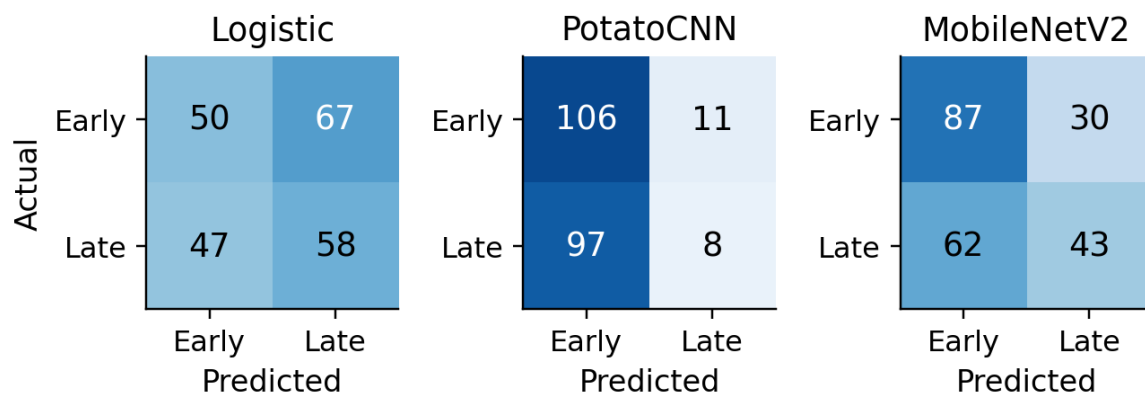


Figure 6. External PlantDoc confusion matrices. Rows are actual labels and columns are predicted labels. Late Blight is the positive class.

MobileNetV2 has the strongest external accuracy, but still misses 62 of 105 Late-Blight images. PotatoCNN misses 97. The external evaluation therefore changes the interpretation of the controlled-domain result.

Internal model comparison

External model metrics

3.8 Robustness, segmentation and ablation

Completed sensitivity experiments on the custom CNN

Condition	Accuracy	Late F1	ROC-AUC
Original colour	0.9533	0.9533	0.9952
Dark	0.4933	0.6607	0.4951
Bright	0.7000	0.5631	0.9761
Contrast	0.8633	0.8783	0.9924
Blur	0.8733	0.8862	0.9884
Rotation	0.9500	0.9492	0.9947
Noise	0.9500	0.9492	0.9934

Darkness causes the strongest loss of accuracy and ranking quality. Brightness preserves relatively high ROC-AUC while reducing F1, indicating that score ranking and a fixed operating threshold can behave differently. These perturbations provide controlled sensitivity evidence; they do not cover all real-world camera and lighting variation.

Segmented representation

Applying the colour model to segmented images at threshold 0.50 produces no positive predictions in the recorded experiment. Selecting a threshold on segmented validation images yields 0.01. Applied once to segmented test images, this gives accuracy 0.8800, F1 0.8784 and ROC-AUC 0.9566. The result shows substantial representation sensitivity. Threshold adjustment is not equivalent to establishing calibrated probabilities.

Validation-only regularisation ablation

Configuration	Validation accuracy	Validation F1
No augmentation; no dropout	0.9967	0.9967
Augmentation; no dropout	0.9933	0.9935
No augmentation; dropout	0.9933	0.9935
Augmentation; dropout	0.9933	0.9935

The ablation outcomes are validation results from separate training runs, not replacements for the frozen main test result. Small differences in this highly separable source do not establish that regularisation is unnecessary. There is no multi-seed confidence interval or external ablation comparison in the supplied exports.

artifacts/tables/robustness_results.csv; segmented_test_metrics.csv;
segmented_calibrated_metrics.csv; ablation_results.csv. Notebook headings: segmented threshold calibration; validation-only ablation study.

3.9 Error analysis, explainability and probability quality

Diagnostics provide evidence for review, with explicit limits

The current custom-CNN error export contains 14 images: nine Early-Blight images predicted Late and five Late-Blight images predicted Early. The older final-report caption referring to “all five” errors is stale. The full 14-row CSV should be the authoritative error inventory, even where an image montage displays only a subset.

Interpretability outputs

The notebook generates predicted-class Grad-CAM overlays and PCA/t-SNE views of learned features. These help inspect model sensitivity, class separation and problematic examples. Heatmaps are not lesion annotations or causal explanations. Feature separation within the internal test set is not evidence of reliable separation on PlantDoc.

Diagnostic	Completed evidence	Interpretive limit
Error inventory	misclassified_samples.csv	An image-level record, not expert diagnosis.
Grad-CAM	gradcam_predicted_class.png	Sensitivity map; not proof of pathology.
Feature projection	feature_pca.png; feature_tsne.png	Exploratory embedding views, not statistical validation.
Reliability	calibration_curve.csv; reliability_diagram.png	Controlled-domain probability assessment.

Confidence and calibration

The stored confidence cell reports a mean confidence of 0.9652, a mean confidence of 0.9764 for correct predictions and 0.7371 for errors, with a Brier score of 0.0299. Confidence is computed from the larger of the two class scores. These summaries describe the recorded custom model on internal data. They do not validate confidence under domain shift.

The optional prototype flags a maximum class score below 0.75. This is an implemented heuristic, not an empirically validated rejection policy. A confidently wrong external prediction may pass it. Prospective work should evaluate calibration, referral frequency and error rates on accepted versus referred cases, using data representative of the intended setting.

Progress implication

The project now offers more than aggregate accuracy: reviewers can trace error direction, inspect example sensitivity and compare confidence with correctness. The next analytical improvement is to connect these diagnostics to a tested data-quality or abstention policy, rather than merely adding more visualisations.

artifacts/tables/misclassified_samples.csv; calibration_curve.csv; feature_embeddings.csv.
Notebook headings: misclassified test images; predicted-class Grad-CAM; confidence + calibration + uncertainty visualisation.

3.9a Robustness and calibration figures

Sensitivity and confidence diagnostics for the custom CNN

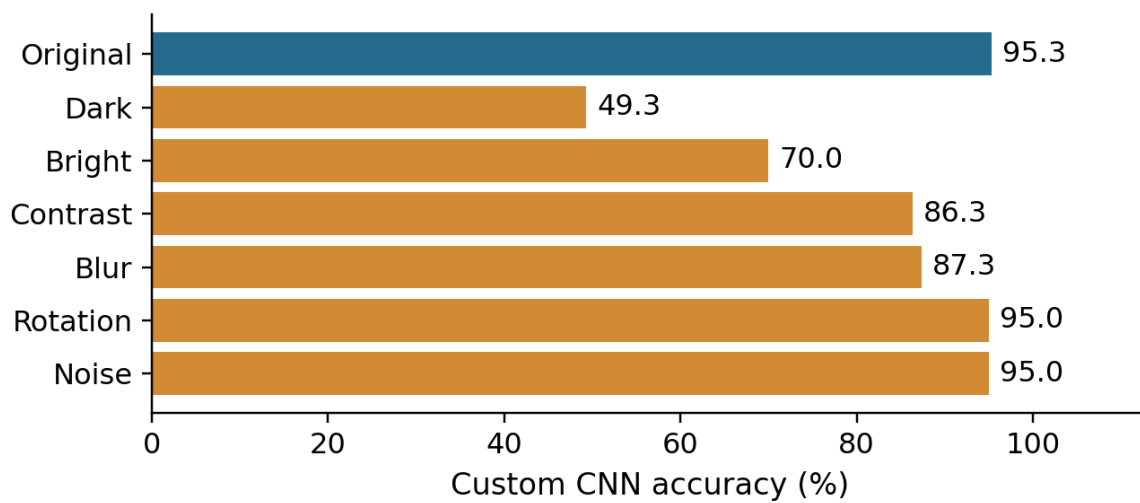


Figure 7. Custom-CNN accuracy under the recorded image perturbations. Darkness produces the largest deterioration. Source: robustness_results.csv.

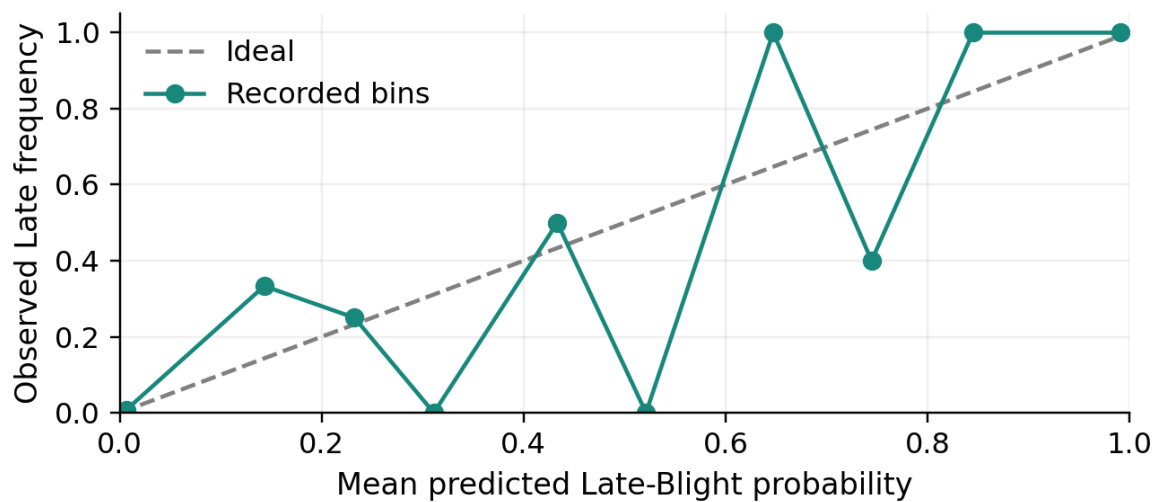


Figure 8. Internal-test reliability diagram from exported probability bins. Departures from the diagonal indicate disagreement between model scores and observed frequency.

The reliability curve is descriptive: bin sample counts are not shown and external-domain calibration is not established. Lighting sensitivity and PlantDoc results remain important limits on the model's confidence.

Perturbation results

Calibration bins

3.10 Workflow, reproducibility and interface progress

Executed notebook available; independent reproduction and release integration remain open

The working sequence is acquisition → inventory/grouping → split → preprocessing → model fitting → validation locking → internal evaluation → external evaluation and diagnostics → packaging. The notebook contains executed outputs for the core analysis and exports evidence used in this report. This review checks consistency of saved evidence and does not claim a new training run.

Recorded runtime item	Value in ENVIRONMENT.txt
Python	3.13.15
PyTorch / Torchvision	2.11.0+cu128 / 0.26.0+cu128
Device / CUDA	CUDA device / 12.8
Seed	42
Custom model profiling	93,377 parameters; 0.3601 MB; 0.8186 ms/image; 1,221.6 images/second

Timing is specific to the recorded runtime and is not a mobile-device latency guarantee. A core environment record exists, but a clean-environment rerun has not been demonstrated. The public repository has a short root README; the inspected public tree contains no notebook, model files or requirements file. The local notebook and archived models are available separately. Expand setup and execution instructions and publish the missing technical files.

Reproduction acceptance procedure — proposed next work

Step	Acceptance evidence
1. Assemble the release	Publish the local notebook and three archived models; expand the public README and add dependency/setup instructions.
2. Recreate the data	Retrieve the recorded revisions, resolve paths and confirm the 2,000-image / 500-group inventory.
3. Run and compare	Execute in a clean environment; compare split hashes, audit counts and exported metrics with the saved reference.
4. Record the result	Save environment, versions, execution log, discrepancies and reviewer sign-off.

Prototype status

The notebook implements a Gradio inference function returning Early/Late scores and a scope warning, with a low-confidence flag. LAUNCH_DEMO is False in the supplied code. The code is evidence of a prototype; it is not evidence of an active deployed service or completed

usability testing. A demonstration should verify model loading and output behaviour before claiming live operation.

artifacts/documentation/ENVIRONMENT.txt; artifacts/tables/model_efficiency.csv;
notebook headings: responsible prediction-interface prototype; corrected report-ready package. Archived models: Potato_Blight_Final_Report_Ready.zip/models/.

Verified repository tree

3.11 Project governance and change management

Technical task records exist; collaboration claims need authentic activity evidence

The local task register lists POT-01 through POT-15 as Done and POT-16 as Needs team input. The new responsibilities record links these allocations to Kavya Sree Sunkara, Sayed Mahidul Islam Rafel, Mahdi Hasan Abdullah, Partha Mallik and Datta Sai Kanaparthi. Allocation establishes expected responsibility; it does not by itself verify completion of every task.

Recorded change	Reason	Evidence / remaining governance action
Mandatory group-disjoint split	Avoid repeated captures of one leaf crossing partitions.	Overlap CSVs exist; attach the decision owner and review link.
Add logistic and MobileNetV2	Broaden the comparison beyond a single CNN.	Comparison and training outputs exist; link implementation commits.
Validation-only decision policy	Keep final test and external labels out of selection.	Notebook and threshold records exist; record peer review.
External and sensitivity testing	Measure generalisation and reliability limits.	PlantDoc and robustness CSVs exist; link board acceptance.
Guarded prototype	Communicate supported classes and uncertainty.	Inference code exists; live demonstration verification remains open.

Project monitoring and sprint evidence

The public repository contains four commits at review. Partha Mallik (Partha115) uploaded assessment reports in commit 22d1031 and 97 architecture, results and planning files in d328570 on 9 September 2026 UTC. Earlier proposal/README commits use the account mahdihasan0017. These establish repository activity, not authorship of every uploaded artefact. No active board URL or individual review records were supplied; the Projects page could not be inspected publicly.

Proposed next milestone sequence

Priority	Work package	Completion criterion
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1	Evidence reconciliation and packaging	Consistent results; publish notebook/models; complete setup and execution instructions.
2	Governance and individual evidence	Project-board URL, member-specific contribution and review evidence; each Lab 1 page complete.
3	Clean reproduction and demonstration	Recorded rerun or explicit unresolved discrepancies; tested prototype or static demonstration.
4	Research extension	Representative field-development plan, class-scope expansion and untouched confirmation protocol.

project_management/task_register.csv; STATUS_BOARD.md; PROJECT_TIMELINE.md; milestones.csv; SCOPE_CHANGE_LOG.md; COLLABORATION_EVIDENCE.md; planning/TASK_ALLOCATION.md. No completion dates are invented in this continuation.

3.12 Risk, ethics, privacy and security

Controls tied to observed limitations and current project scope

Risk	Evidence / consequence	Current control and next action
Domain shift — high	MobileNetV2 accuracy falls from 99.67% to 58.56%; unreliable transfer.	Retain experimental scope; plan representative, expert-reviewed evaluation.
Unsupported input — high	No Healthy, Other or Unknown class; forced binary output.	Visible two-class warning; develop and validate rejection/expanded-class handling.
Leakage — high	Four captures per leaf create dependence and splitting risk.	Group isolation and hash audits; retain group IDs and use group-aware uncertainty.
Overconfidence — high	High internal confidence does not survive external shift.	Probability warning and heuristic flag; validate calibration and referral policy.
Packaging / reproducibility — medium	Model paths in task register do not resolve in extracted repository.	Models confirmed in archive; integrate them and test clean reproduction.

Evidence / participation — high	Role assignments exist; evidence remains incomplete for several members.	Require authentic member statements, lab evidence, issues and reviews.
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Ethical scope and reporting

The system predicts one of two disease labels and does not establish a diagnosis or recommend treatment. Report class-specific failures and external results alongside headline internal performance. Do not infer that public image labels or Grad-CAM highlights constitute expert confirmation. The proposal's healthy-class wording is corrected in this continuation.

Privacy and data stewardship

The supplied dataset is plant imagery, and no user-upload service is evidenced as active. If future collection includes farm images, location metadata, people or participant details, establish a defined purpose, consent where appropriate, access controls and retention/deletion rules. The current project should preserve provenance and resolve the licensing questions recorded in its source notes.

Security and operational boundary

A future image-upload service would need input validation, file-size limits, dependency maintenance, controlled access to secrets and restricted retention. These controls are proposed requirements, not completed features. Research checkpoints and joblib files should be loaded only from trusted project sources. Keep deployment claims separate from notebook execution evidence.

planning/RISK_REGISTER.csv; artifacts/tables/risk_register.csv;
artifacts/documentation/SCIENTIFIC_LIMITATIONS.txt;
DATA_PROVENANCE_AND_LICENSES.txt;
GOVERNANCE_ACTIONS_REQUIRED.txt.

3.13 Evaluation of progress and remaining work

Assessment against the original objectives and current evidence

Objective	Current assessment	Evidence / outstanding work
Acquire and organise data	Implemented	Pinned inputs, inventory, class counts and group-aware partitions.
Preprocess images	Implemented	Deterministic evaluation transforms and training-only augmentation.
Develop CNN classifier	Implemented and extended	PotatoCNN, traditional baseline and transfer comparator; models in archive.

Evaluate performance	Implemented and extended	Internal/external metrics, errors, thresholds and sensitivity experiments.
Visualise and critique	Implemented	Figures, calibration, explainability and explicit domain limitations.
Repository and collaboration	Partial	Named roles and a public repository exist; publish notebook/models and complete member-specific evidence.
Individual assessment pages	Incomplete	Five named pages prepared; screenshots, member-specific contributions and reflections remain required.

The milestone judgement remains substantial technical progress with incomplete individual assessment evidence. The new files resolve team identity and planned responsibility, and repository inspection confirms public upload activity. Technical results support a working controlled-domain classifier; external performance still limits deployment claims.

Corrections applied in this progress report

External evaluation and robustness are reported as completed because current exports exist, despite older slide claims that they are pending. The custom model's 14 errors replace stale five-error wording. Thresholds 0.50 and 0.38 have the same accuracy but different precision/recall and decisions. The model archive is distinguished from missing extracted repository model paths. Planned roles are not presented as verified individual contributions.

Immediate completion criteria

Before submission, complete all five named students' laboratory screenshots, personal contribution statements and reflections. Add a usable project-board URL and member-specific evidence links. Publish the notebook and archived models, complete reproduction instructions and check the final searchable PDF remains within 30 pages. Names, IDs, role allocations and the repository URL have now been filled from the new source files.

Basis: original proposal objectives; notebook outputs; final_model_comparison.csv; plantdoc_external_metrics.csv; validation_threshold_comparison.csv; task_register.csv; local repository and ZIP inventories.

3.14 Assessment compliance and evidence index

Submission preparation status and source register

Assessment requirement	Status after this update
Section 1: complete Assessment 1	Newly supplied team-specific four-page proposal preserved unchanged.
Section 2: one-page update	One page with named roles and a clickable repository link.
Section 3: technical continuation	18 pages, including four figure pages and seven new figures.
Section 4: one page per student	Five named pages with student IDs and documented responsibilities.
Single searchable PDF; maximum 30 pages	28 pages total; original figure plus seven numbered continuation figures.
Text at least 10 points	New text, table text and figure labels designed at 10 points or larger.
Project / individual evidence links	Repository, result artefacts and observed upload commits linked; board URL and further individual evidence remain required.
Laboratory and reflections	Lab 1 screenshots and personal reflections still missing.

Evidence register

Local sources: PRT661_Assessment_2_No_Duplicates.pdf (assessment rules); Dan Group 7 Theme 2 PP.pdf (Assessment 1, identities and IDs); team_structure_responsibilities.pdf (named allocations); CODE.ipynb (same implementation as the prior notebook); exported CSVs, figures, models and planning records described throughout this report.

Online sources: the public repository and GitHub commit records were inspected on 10 September 2026. The inspected main-tree revision is d3285704476a7f1ef19728486539cee4f6b29b30. Links to saved result files use that revision so the evidence remains tied to the reviewed state.

Public repository and README

Reviewed repository tree

Commit history

Figure register

Figure 1: original proposal architecture (Section 1). Figure 2: updated architecture. Figure 3: image and leaf-group splits. Figure 4: training histories. Figure 5: internal/external accuracy. Figure 6: external confusion matrices. Figure 7: perturbation accuracy. Figure 8: reliability diagram.

Figures 2–8 were prepared from the supplied architecture specification and CSV results; no model was retrained. PNG and vector PDF versions are included in the figures folder. The editable DOCX contains Sections 2–4; the consolidated PDF includes the unchanged Assessment 1 report.

Section 4: Individual Work and Contribution

4.1 Kavya Sree Sunkara | S386025

Danala Group 7 | Data acquisition and repository setup

Laboratory 1 evidence

Lab: Querying Data by Using Athena (Module 4). Recorded score: 45/45 (100.00%) is verified from the available evidence.

The screenshot shows the AWS Academy interface for the 'Querying Data by Using Athena' lab. The user is logged in as 's386025@students.cdu.edu.au'. The lab title is 'Querying Data by Using Athena' (Version 1.0.21). The lab overview and objectives are displayed, including a story about Sofia and Mary, and the goals of the lab. A table on the right shows the completion status of various tasks.

Task	Score
Total score	45/45
[Task 1a] Database created	5/5
[Task 1b] Database table created	5/5
[Task 1c] Database table previewed	5/5
[Task 2] Database table jan exists	5/5
[Task 4] Student queried using views	5/5
[Task 5a] Student tested the query in Athena	5/5
[Task 5b] CloudFormation stack created	5/5
[Task 5c] CreateNamedQuery event recorded	5/5
[Task 7] GetNamedQuery event recorded	5/5

Individual project contribution

Kavya's documented contribution area is data acquisition and repository setup (POT-01-02). This component covers organising the two-class PlantVillage dataset, recording source provenance and preparing a traceable data inventory. The project outputs include 2,000 images, balanced class counts and documented source revisions, which provide the foundation for reproducible preprocessing and model development.

Supporting evidence

[Data provenance](#) | [Class inventory](#)

Professional reflection

The principal lesson from this component is that dataset traceability is part of model quality. Recording source revisions, labels and storage locations makes later results easier to audit. Future work should maintain a clear connection between each dataset change and its supporting repository record.

Section 4: Individual Work and Contribution (continued)

4.2 Sayed Mahidul Islam Rafel | S387949

Danala Group 7 | Image preprocessing and exploratory analysis

Laboratory 1 evidence

Lab: Querying Data by Using Athena (Module 4). Recorded score: 45/45 (100.00%). The session ended on 2 September 2026 at 23:39:14 (UTC-07:00). The screenshots identify S387949; they show a graded attempt rather than full marks. The end time is session evidence, not a separate grading timestamp.

The top screenshot displays the AWS Academy lab interface for 'Querying Data by Using Athena'. The left sidebar shows the user's account (s387949@students.cdu.edu.au) and navigation options. The main content area shows the lab overview and objectives, including a description of the task and a list of tasks to be completed. The right sidebar shows a task list with a total score of 45/45.

The bottom screenshot shows the same lab interface, but with a 'Cloud Labs' summary panel on the right. This panel displays session details: Session started at: -0001-11-30T00:00:00-0752, Session ended at: 2026-09-02T23:39:14-0700, and Accumulated lab time: 04:00:00(240 minutes).

Individual project contribution

Sayed's documented contribution area covers preprocessing and exploration (POT-03-05): leaf-group construction, split design and training-only augmentation. The resulting evidence records 350/75/75 leaf groups across training, validation and testing, with no shared groups. These controls reduce leakage between related images.

Supporting evidence

[Group-overlap audit](#) | [Split summary](#)

Professional reflection

The key lesson is that related images must stay in the same partition. Data preparation affects the credibility of evaluation, not just image format. The 45/45 lab result also identifies scope to revisit uncredited tasks and verify the final assessment evidence.

Section 4: Individual Work and Contribution (continued)

4.3 Mahdi Hasan Abdullah | S394885

Danala Group 7 | Model development and training

Laboratory 1 evidence

Lab: Querying Data by Using Athena (Module 4). The screenshot identifies S394885 and records the session ending on 29 August 2026 at 13:31:17 (UTC+07:00), with 131 minutes of accumulated lab time. Recorded score: 45/45 (100.00%).

The screenshot shows the AWS Academy interface for the 'Querying Data by Using Athena' lab. The sidebar on the left contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area displays the lab title, overview, and objectives. On the right, a 'Total score' of 45/45 is shown, along with a list of tasks and their completion status.

Task	Status
[Task 1a] Database created	5/5
[Task 1b] Database table created	5/5
[Task 1c] Database table previewed	5/5
[Task 2] Database table jan exists	5/5
[Task 4] Student queried using views	5/5
[Task 5a] Student tested the query in Athena	5/5
[Task 5b] CloudFormation stack created	5/5
[Task 5c] CreateNamedQuery event recorded	5/5

The screenshot shows the AWS Academy interface for the 'Querying Data by Using Athena' lab. The sidebar on the left contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area displays the lab title, overview, and objectives. On the right, a 'Cloud Labs' section shows session details, including the start and end times, and the accumulated lab time.

Cloud Labs
Session started at: -0001-11-30T00:00:00-0752
Session ended at: 2026-08-29T13:31:17-0700
Accumulated lab time: 02:11:00(131 minutes)

Individual project contribution

Mahdi's documented contribution area is model development and training (POT-06-08), covering the logistic baseline, custom PotatoCNN and MobileNetV2 comparison. The recorded implementation includes early stopping, saved model weights and validation-selected thresholds. The custom CNN has 93,377 parameters, while MobileNetV2 provides the strongest internal accuracy at 99.67%.

Supporting evidence

[CNN configuration](#) | [Model comparison](#) | [Training history](#)

Professional reflection

The model comparison shows why a high internal score needs an external check. MobileNetV2's accuracy falls to 58.56% on PlantDoc, so further progress depends on representative data as well as architecture choice. Checkpoint selection and threshold decisions should remain separate from final testing.

Section 4: Individual Work and Contribution (continued)

4.4 Partha Mallik | S392819

Danala Group 7 | Evaluation, visualisation and error analysis

Laboratory 1 evidence

Lab: Querying Data by Using Athena (Module 4). Recorded score: 45/45 (100%). The session ended on 3 September 2026 at 12:36:03 (UTC+07:00). Both screenshots identify S392819. The full score supports completion of the graded tasks; the displayed timestamp records the session end.

The image displays two screenshots of the AWS Academy lab interface. Both screenshots show the user's account information (s392819@students.cdu.edu.au) and the lab title 'Querying Data by Using Athena'. The left screenshot shows the 'Grades' tab, which lists the total score as 45/45 and provides a breakdown of scores for various tasks. The right screenshot shows the 'Cloud Labs' tab, which displays session details including the start and end times and the accumulated lab time.

Task	Score
[Task 1a] Database created	5/5
[Task 1b] Database table created	5/5
[Task 1c] Database table previewed	5/5
[Task 2] Database table jan exists	5/5
[Task 4] Student queried using views	5/5
[Task 5a] Student tested the query in Athena	5/5
[Task 5b] CloudFormation stack created	5/5
[Task 5c] CreateNamedQuery event recorded	5/5

Cloud Labs
Session started at: -0001-11-30T00:00:00-0752
Session ended at: 2026-09-03T12:36:03-0700
Accumulated lab time: 10:18:00(618 minutes)

Individual project contribution

Partha's documented area covers evaluation, visualisation and error analysis (POT-09–14). Public commits under Partha115 record uploads of assessment reports and the architecture, results and planning package. These provide direct evidence of repository integration alongside the project's internal/external evaluation outputs.

Supporting evidence

[Visualization & Error Analysis](#) | [Report upload: 22d1031](#) | [Evidence upload: d328570](#)

Professional reflection

Evaluation must make limitations visible as well as report accuracy. The 14 custom-CNN errors and the PlantDoc performance gap demonstrate the importance of class-specific analysis and traceable result records. Future reviews should connect each conclusion to its prediction table and documented decision.

Section 4: Individual Work and Contribution (continued)

4.5 Datta Sai Kanaparthi | S385673

Danala Group 7 | Documentation, ethics review and final integration

Laboratory 1 evidence

Lab: Querying Data by Using Athena (Module 4). Recorded score: 45/45 (100.00%).

The screenshot shows the AWS Academy interface for the 'Querying Data by Using Athena' lab. The user is logged in as 's385673@students.cdu.edu.au'. The lab overview and objectives are displayed, along with a table of tasks and their completion status.

Task	Score
Total score	45/45
[Task 1a] Database created	5/5
[Task 1b] Database table created	5/5
[Task 1c] Database table previewed	5/5
[Task 2] Database table jan exists	5/5
[Task 4] Student queried using views	5/5
[Task 5a] Student tested the query in Athena	5/5
[Task 5b] CloudFormation stack created	5/5
[Task 5c] CreateNamedQuery event recorded	5/5
[Task 7] GetNamedQuery event recorded	5/5

Individual project contribution

Datta's documented contribution area is documentation, ethics review and final integration (POT-15–16). This area brings the project objectives, architecture, experimental results, risk analysis and governance records into a coherent report. Its central requirement is to keep the claims consistent with the evidence, including the two-class scope and the limitations of external performance.

Supporting evidence

[Scope-change record](#) | [Risk register](#) | [Assessment records](#)

Professional reflection

Clear documentation helps readers distinguish implemented work from proposed extensions. The external results and unresolved evidence gaps show why a report should be explicit about limitations and version history. Future integration should reconcile every figure, table and claim with the final recorded experiment.